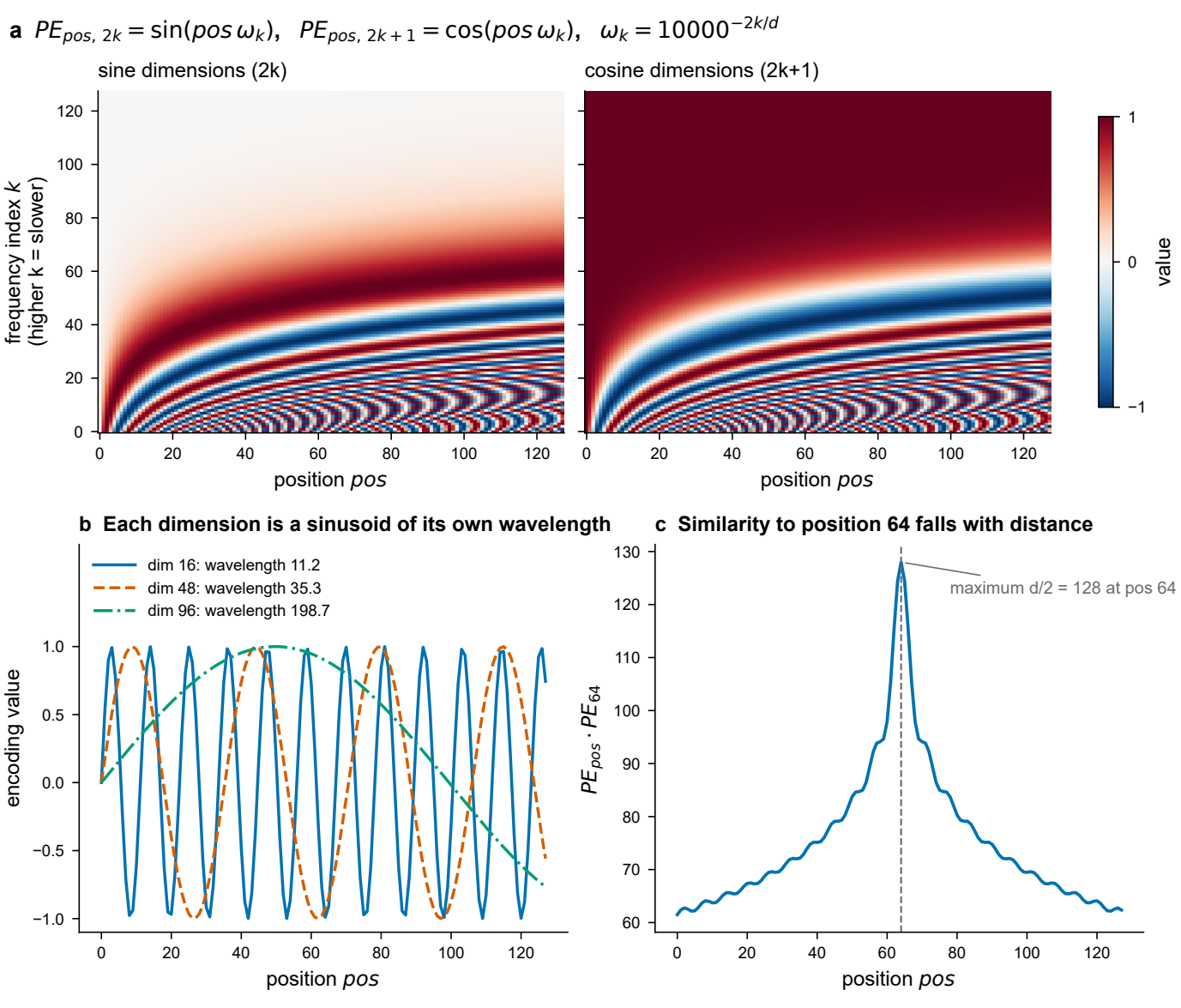

a $PE_{pos,\,2k} = \sin(pos\,\omega_k)$, $PE_{pos,\,2k+1} = \cos(pos\,\omega_k)$, $\omega_k = 10000^{-2k/d}$
sine dimensions (2k)
cosine dimensions (2k+1)
frequency index k
(higher k = slower)
position pos
position pos
value
1
0
−1
b Each dimension is a sinusoid of its own wavelength
dim 16: wavelength 11.2
dim 48: wavelength 35.3
dim 96: wavelength 198.7
encoding value
position pos
c Similarity to position 64 falls with distance
maximum d/2 = 128 at pos 64
$PE_{pos} \cdot PE_{64}$
position pos